

Reconstruction from a Bounded Observer State

Compression and recovery in Observer-Patch Holography (OPH)

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Abstract

In OPH no global computer renders 3D particle coordinates frame by frame. What we perceive as three-dimensional space, fields and time moving forward are observer-facing readouts of a computed quotient normal form on a holographic screen. The substrate has no background clock, no 3D space and no master copy of the world. It has a finite network of observer patches, a mismatch score on every shared boundary seam and a local repair move that lowers those scores, and each repair move is a quantum recovery channel. A bounded observer therefore needs no copy of the world beyond its boundary. This note makes that quantitative: the relational information that can be removed from an observer’s representation is measured by the conditional mutual information across its boundary collar, which bounds the best reconstruction error from above and below.

1. Rendering without a master copy

Bulk spacetime is read off the repaired normal form. The geometry readout is a function of that gauge-invariant quotient and does not depend on hidden carrier coordinates, gauge representatives or the repair schedule [SM, Thm. 6.56]. On its declared branch the spacetime paper recovers flat 1+3 causal order, Lorentz-frame kinematics and, under further stated premises, the Einstein equation up to a cosmological constant [ST]. There is no master database holding the state of every atom. The computation can be split over any number of workers and shards, and every admissible execution gives the same observer readouts whatever the partition, schedule or restart history [C, Thm. 9.5, Cor. 9.6]. An observer holds a finite patch whose records form a commutative central algebra, and its local world is evaluated when it reads that algebra [C, Thm. 7.2].

Stefaan Vossen asked, in a discussion of compression and what he calls holographic beingness, how much relational information can be removed from a representation before an independent operator can no longer reconstruct the relevant larger state. OPH answers this with the collar Markov condition and its Petz/Fawzi–Renner recovery channels (Section 3). The repair move of the substrate is itself such a recovery channel, so the one update rule of an OPH world is reconstruction from a compressed boundary. Vossen’s slogan “understanding is reconstruction from compressed representation” is, in OPH, the dynamical law.

2. The substrate

Patch net [C, Def. 2.1–2.2, Prop. 2.3]. A finite connected graph $G = (V, E)$; patch i has a finite state space S_i , the global space is $\Sigma = \prod_i S_i$, and each overlap $e = \{i, j\}$ has an interface alphabet I_e with restriction maps $\pi_{i,e} : S_i \rightarrow I_e$. The mismatch potential

$$\Phi(s) = \sum_{e=\{i,j\} \in E} w_e d_e(\pi_{i,e}(s_i), \pi_{j,e}(s_j)), \quad w_e > 0, \quad d_e(a, b) = 0 \iff a = b,$$

vanishes exactly on the globally consistent set $C = \Phi^{-1}(0)$, a finite constraint code.

Repair [C, Def. 3.1–3.2]. Patch i carries a collar chart $A_i - B_i - D_i$ around its overlaps. A repair move T_i changes only patch i and is induced by a recovery move on that collar: the exact Markov splice when $I(A_i : D_i | B_i) = 0$, otherwise $(\text{id}_{A_i} \otimes \mathcal{R}_i)(\omega_{A_i B_i})$ with $\mathcal{R}_i$ a Petz/Fawzi–Renner recovery channel. A move is accepted only if it strictly lowers the score Φ_i of the overlaps it touches.

Normal form [C, Thm. 3.25, Cor. 3.26, Thm. 5.2]. Termination, local confluence on the gauge quotient $\underline{Q} = \Sigma/\Gamma$, repair completeness and Newman’s lemma give each initial state s a unique quotient normal form $\text{nf}(q(s)) \in q(C)$, reached by every maximal asynchronous repair run. Every quotient observable takes the same value for every update order, and gauge-equivalent inputs reach the same normal form. Bare consensus fixes no metric by itself [C, Thm. 2.6]; geometry enters through the declared readout branch of [SM, ST].

Pairwise agreement is not enough [C, Thm. 4.1, Cor. 4.2]. For labels in an abelian group, the seam equations $x_v - x_u = b_e$ have a global solution if and only if $\sum_{e \in \gamma} \varepsilon_\gamma(e) b_e = 0$ on every cycle γ . On a $\mathbb{Z}_2$ triangle with $b = (0, 0, 1)$ each seam is satisfiable on its own and the triangle is not. In the language of tomography, one projection never suffices, and global consistency is a condition on cycles of views.

Records [C, Def. 7.1, Thm. 7.2]. On a completed slice with accessible algebra $\mathcal{A}_t$, the record projectors $\widehat{P}_a(t) \in Z(\mathcal{A}_t)$ generate a finite commutative algebra $\mathcal{Z}_{\text{rec}}(t)$. For an event E in it, $\mathbb{P}_t(E) = \text{Tr}(\rho_t \widehat{P}_E)$, the declared Lüders update is $\rho_t|_E = \widehat{P}_E \rho_t \widehat{P}_E / \text{Tr}(\rho_t \widehat{P}_E)$, and an immediate reread returns E with probability one.

3. Removal and reconstruction

Let A be the patch interior, B its collar and D everything beyond. The observer holds ρ_{AB} ; the larger state is ρ_{ABD} . With von Neumann entropies S in nats and modular Hamiltonians $K_X = -\ln \rho_X$,

$$I(A:D | B) = S(AB) + S(BD) - S(B) - S(ABD) = \text{Tr} \rho_{ABD} (K_{AB} + K_{BD} - K_B - K_{ABD}) \geq 0$$

[LR; C, App. A]. It measures the correlation between A and D that B does not carry. Write

$$\delta_\star := \min_{\mathcal{R}: B \rightarrow BD} \frac{1}{2} \|\rho_{ABD} - (\text{id}_A \otimes \mathcal{R})(\rho_{AB})\|_1$$

for the best reconstruction error of an operator that acts on the collar alone.

Exact case. $I(A:D | B) = 0$ if and only if $\rho_{ABD} = (\text{id}_A \otimes \mathcal{P})(\rho_{AB})$ for the Petz map [P1, P2]

$$\mathcal{P}_{B \rightarrow BD}(X) = \rho_{BD}^{1/2} (\rho_B^{-1/2} X \rho_B^{-1/2} \otimes 1_D) \rho_{BD}^{1/2} \quad (\text{inverses on supports}),$$

and if and only if $\mathcal{H}_B = \bigoplus_\alpha \mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha}$ with [HJPW]

$$\rho_{ABD} = \bigoplus_\alpha p_\alpha \rho_{Ab_L^\alpha}^{(\alpha)} \otimes \rho_{b_R^\alpha D}^{(\alpha)}.$$

On this form the splice identity of [C, Thm. A.1] holds: replacing each $\rho_{b_R^\alpha D}^{(\alpha)}$ by any state $\sigma_{b_R^\alpha D}^{(\alpha)}$, on the same sectors leaves $\text{Tr}(X\rho)$ unchanged for every observable X on $A \cup b_L$. The entire exterior can be removed and exchanged; what must be kept is the sector structure of the collar.

Approximate case. For every state, with $d_A = \dim \mathcal{H}_A$ and h the binary entropy in nats,

$$\delta_\star^2 \leq 1 - e^{-I(A:D|B)} \leq I(A:D | B), \quad (1)$$

$$I(A:D | B) \leq 2\delta_\star \ln d_A + (1 + \delta_\star) h\left(\frac{\delta_\star}{1+\delta_\star}\right). \quad (2)$$

Bound (1) is the Fawzi–Renner theorem $I(A:D | B) \geq -2 \ln F(\rho_{ABD}, (\text{id}_A \otimes \mathcal{R})(\rho_{AB}))$, $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$ [FR], combined with $\frac{1}{2}\|\rho - \sigma\|_1 \leq \sqrt{1 - F^2}$ [FvdG]; the channel $\mathcal{R}$ can be taken universal, built from ρ_{BD} alone [JRSWW, Cor. 4.1]. For (2), let $\sigma = (\text{id}_A \otimes \mathcal{R})(\rho_{AB})$ attain $\delta_\star$. Data processing gives $I(A:BD)_\sigma \leq I(A:B)_\rho$ and $\sigma_A = \rho_A$, so $I(A:D | B)_\rho = I(A:BD)_\rho - I(A:B)_\rho \leq S(A|BD)_\sigma - S(A|BD)_\rho$, which Winter’s continuity bound controls [W, Lemma 2]. Bounded observables inherit (1): $|\text{Tr} X(\rho - \sigma)| \leq 2\|X\|_\infty \delta_\star$.

Removal criterion. Let the observer keep only a part B' of its boundary representation, $B = B'B''$. Rebuilding the dropped part together with the exterior is the same problem with $B \rightarrow B'$ and $D \rightarrow B''D$. For a target error η , an operator acting on B' alone succeeds whenever

$$I(A:B''D | B') \leq -\ln(1 - \eta^2),$$

and by (2) no operator on B' succeeds once $I(A:B''D | B') > 2\eta \ln d_A + (1 + \eta) h(\eta/(1 + \eta))$. The relational information that can be removed is the part whose removal keeps this conditional mutual information below tolerance. The decoding rule is fixed by $\rho_{B'B''D}$, the statistics of the world beyond the interior, so the observer needs no copy of D , only access to the decoding rule.

Which boundary factor faces the observer. Zero conditional mutual information fixes the split of B from the state. The geometric statements of OPH need that split to agree with the preselected edge split of the collar, a separate condition called Markov-split alignment [ST, Sec. 2.3]. Four qubits with $\rho = \Phi_{AB_R} \otimes \Phi_{B_L D}$ (Bell pairs across the cut) have $I(A:D | B) = 0$ and $I(A:B_R) = 2 \ln 2$: the compression is exact, and the factor of B that faces the observer is B_R . Storage and representation separate already at the level of a single collar.

What the part carries about the whole. For a gauge-invariant collar with sectors α of dimension d_α , the one-sided state obeys [FL, Thm. 5.2]

$$S(\rho_L) = H(p) + \sum_\alpha p_\alpha S(\rho_{L,\alpha}) + \sum_\alpha p_\alpha \ln d_\alpha,$$

and the last term is the expectation of the central observable $Z_L = \sum_\alpha (\ln d_\alpha) P_\alpha^L$. The boundary imprint sits in the commutative center, the same place as the records.

4. Compression vocabulary in OPH

larger state	ρ_{ABD}
storage	ρ_{AB} : the patch and its collar
representation	collar sectors α and records in the commutative center
decoding rules	recovery channel $\mathcal{R}_{B \rightarrow BD}$, fixed by ρ_{BD}
reconstruction	$(\text{id}_A \otimes \mathcal{R})(\rho_{AB})$, error bounded by (1)–(2)
what can be removed	all of D , and any B'' with $I(A:B''D B') \approx 0$
additional perspectives	overlapping patches; consistency is a cycle condition, and repair gives all of them one normal form

In OPH the relation between a part and the whole is fixed by recoverability, and the whole that observers share is the quotient normal form, which no single patch stores. Vossen’s weaker proposition, that a bounded being carries recoverable relations to a whole it never becomes, holds there with the conditional mutual information as its measure.

References

OPH papers, paper release r2043 (13 September 2026):

- [C] B. Mueller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*, https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/reality_as_consensus_protocol.pdf.
- [SM] B. Mueller, A. Osika, K. Xue, B. Cassie, B. de La Fournière, *Federated Echoshedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH*, https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/screen_microphysics_and_observer_synchronization.pdf.
- [ST] B. Mueller, A. Osika, M. Ponder, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, B. de La Fournière, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*, https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf.
- [FL] B. Mueller, A. Osika et al., *Finite Observer Consensus as a Reconstruction Principle: Normal Forms, the Standard Model Lie Type, and a Route to the Einstein Field Equation* (flagship paper), <https://philpapers.org/rec/MUEFOC>; https://github.com/FloatingPragma/observer-patch-holography/blob/main/flagship/from_observer_consensus_to_standard_physics.pdf.

Quantum information:

- [FR] O. Fawzi, R. Renner, Quantum conditional mutual information and approximate Markov chains, *Commun. Math. Phys.* **340**, 575–611 (2015).
- [FvdG] C. A. Fuchs, J. van de Graaf, Cryptographic distinguishability measures for quantum-mechanical states, *IEEE Trans. Inf. Theory* **45**, 1216–1227 (1999).
- [HJPW] P. Hayden, R. Jozsa, D. Petz, A. Winter, Structure of states which satisfy strong subadditivity of quantum entropy with equality, *Commun. Math. Phys.* **246**, 359–374 (2004).
- [JRSWW] M. Junge, R. Renner, D. Sutter, M. M. Wilde, A. Winter, Universal recovery maps and approximate sufficiency of quantum relative entropy, *Ann. Henri Poincaré* **19**, 2955–2978 (2018).
- [LR] E. H. Lieb, M. B. Ruskai, Proof of the strong subadditivity of quantum-mechanical entropy, *J. Math. Phys.* **14**, 1938–1941 (1973).
- [P1] D. Petz, Sufficient subalgebras and the relative entropy of states of a von Neumann algebra, *Commun. Math. Phys.* **105**, 123–131 (1986).
- [P2] D. Petz, Sufficiency of channels over von Neumann algebras, *Quart. J. Math.* **39**, 97–108 (1988).
- [W] A. Winter, Tight uniform continuity bounds for quantum entropies: conditional entropy, relative entropy distance and energy constraints, *Commun. Math. Phys.* **347**, 291–313 (2016).